

SCI225 Week 5 — Complete Study Guide

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CNS & PNS Disorders · Neuromuscular Junction · Pediatric Neurologic Conditions

Study Spine — for every disorder ask:

1. Where is the lesion (CNS / PNS / NMJ)? 2. What is the mechanism? 3. Acute or chronic? 4. What's the time-critical action?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

- **LO1** — Discuss the disorders of the Central and Peripheral Nervous System (Units 1–4)
- **LO2** — Examine the Neuromuscular Junction and why it is important to Pathophysiology (Unit 5)
- **LO3** — Understand potential alterations of neurologic function in children (Unit 6)

Every practice question in this guide is original — written to help you apply the concept, not to reproduce exam items.

Unit 1 • Stroke & Cerebrovascular Disease LO1

1.1 Ischemic vs Hemorrhagic Stroke

	Ischemic (~85%)	Hemorrhagic (~15%)
Mechanism	Vessel blocked → no blood downstream	Vessel ruptures → blood floods brain tissue
Subtypes	Thrombotic (atherosclerotic), embolic (often AFib), lacunar (small penetrating vessels)	Intracerebral (HTN), subarachnoid (aneurysm)
Risk factors	HTN, AFib, atherosclerosis, diabetes, smoking, hyperlipidemia, obesity, prior TIA	HTN (intracerebral), berry aneurysm/ AVM (subarachnoid), anticoagulant use
Onset	Sudden focal deficit; may have warning TIA; thrombotic often awakens with deficit	Sudden severe headache, vomiting, possible loss of consciousness
Imaging	CT often normal early; MRI shows infarct hours later; CT must be done to <i>exclude</i> bleed before tPA	CT shows blood immediately
Treatment	tPA within 3–4.5 h of last known well; mechanical thrombectomy up to 24 h in select patients	tPA contraindicated ; control BP, reverse anticoagulation, surgical decompression if needed

ACRONYM / MNEMONIC FAST for stroke recognition:

Face drooping · Arm weakness · Speech difficulty · Time to call 911

"Time is brain" — every minute, ~2 million neurons die during ischemic stroke.

1.2 The Ischemic Cascade — Why Neurons Die

Vessel occlusion → ↓ O₂ + glucose → ↓ ATP → Na/K pump fails → cell swelling + depolarization → excessive glutamate release → Ca²⁺ influx → activation of destructive enzymes → free-radical injury → neuron death

WHY The primary reason neurons die during a stroke is **energy failure** — without ATP, cells cannot maintain ion gradients. The neuron depolarizes, dumps glutamate, and gets flooded with calcium. This is "excitotoxicity." The penumbra (the area around the core infarct) is salvageable if perfusion is restored quickly — that's the window tPA targets.

PATHOLOGY Stroke risk factors students must know cold:

- **Atrial fibrillation** — clots form in left atrium, travel to brain (embolic stroke)
- **Hypertension** — accelerates atherosclerosis and is the #1 modifiable risk factor
- **Diabetes** — accelerates vascular disease
- **Smoking** — endothelial dysfunction, hypercoagulability
- **Hyperlipidemia / atherosclerosis** — plaque rupture and embolism
- **Prior TIA** — major warning sign; ~10–15% risk of stroke in following 90 days
- **Carotid stenosis**, sickle cell disease, oral contraceptives + smoking

★ **EXAM TIP** TIA = "warning stroke." Same mechanism as ischemic stroke but symptoms resolve completely (typically < 1 hour, by definition < 24 h). Treat aggressively — antiplatelet, statin, BP control, AFib management — to prevent the next event.

Practice Questions — Stroke

Q1. A 70-year-old with atrial fibrillation suddenly develops right arm weakness and slurred speech. Which mechanism is most likely?

- A. Subarachnoid hemorrhage from berry aneurysm
- B. Embolic ischemic stroke from a left atrial clot
- C. Hypertensive intracerebral hemorrhage
- D. Brain tumor

Show answer

B. AFib + sudden focal deficit = embolic stroke until proven otherwise. Atrial stasis allows clot formation; clot embolizes to a cerebral artery. Anticoagulation (warfarin or DOAC) prevents this in AFib.

Q2. What is the primary reason neurons begin to die during a stroke?

- A. Bacterial infection of the brain
- B. Loss of ATP causes ion-pump failure, excitotoxic glutamate release, and calcium-mediated cell injury
- C. The brain becomes too acidic from increased breathing
- D. Antibodies attack neurons

[Show answer](#)

B. Ischemia → energy failure → ion gradient collapse → excitotoxicity → Ca^{2+} influx → enzyme activation → cell death. Restoring perfusion quickly is the only way to save the penumbra.

Q3. Select all that apply. Which findings are associated with increased stroke risk?

- A. Atrial fibrillation
- B. Hypertension
- C. Diabetes mellitus
- D. Daily moderate exercise
- E. Carotid artery stenosis
- F. Prior transient ischemic attack

[Show answer](#)

A, B, C, E, F. D is protective. The other five are major modifiable or addressable risk factors.

Q4. A patient awakens with a unilateral facial droop, arm weakness, and slurred speech. Last seen normal 7 hours ago. Which intervention is most appropriate first?

- A. Administer tPA immediately
- B. Obtain a non-contrast CT head to exclude hemorrhage and assess eligibility for thrombolysis or thrombectomy
- C. Begin therapeutic anticoagulation
- D. Discharge with antiplatelet therapy

[Show answer](#)

B. Always image first to exclude hemorrhage. tPA window is 3–4.5 h from last known well — this patient is past the window for tPA but may still be a thrombectomy candidate up to 24 h with eligible imaging.

Unit 2 • Other CNS Disorders Lo1

2.1 Multiple Sclerosis

MECHANISM MS is a chronic autoimmune demyelinating disease of the **CNS**. T cells attack **oligodendrocytes**, stripping myelin from axons in the brain, spinal cord, and optic nerves. Lesions accumulate in different locations over time → "**dissemination in time and space.**"

Classic MS features:

- Optic neuritis (painful monocular vision loss) — often the first symptom
- Sensory symptoms: numbness, tingling, Lhermitte's sign (electric shock with neck flexion)
- Motor: weakness, spasticity, hyperreflexia, Babinski (UMN signs)
- Cerebellar: ataxia, intention tremor, scanning speech
- Bladder dysfunction, fatigue, cognitive changes
- Heat intolerance (Uhthoff phenomenon — symptoms worsen with elevated body temperature)
- Relapsing-remitting most common pattern; primary or secondary progressive in some

★ **EXAM TIP** "Young woman with episodes of vision loss and limb weakness, with a normal neuro exam between episodes" = MS until proven otherwise. F:M ratio is roughly 3:1; onset typically 20–40.

2.2 Parkinson's Disease

MECHANISM Loss of dopamine-producing neurons in the **substantia nigra** (part of the basal ganglia). Dopamine normally facilitates voluntary movement; without it, the basal ganglia over-inhibit motor cortex output → **bradykinesia and rigidity**.

ACRONYM / MNEMONIC TRAP — Parkinson's cardinal features:

Tremor (resting, "pill-rolling," 4–6 Hz)

Rigidity (cogwheel)

Akinesia / bradykinesia (slow movement initiation)

Postural instability (falls, shuffling gait, stooped posture)

Plus: mask-like face, micrographia, hypophonia

WHY Why levodopa works. Dopamine itself can't cross the BBB. Levodopa (L-DOPA) does, and is converted to dopamine in the brain. It's combined with carbidopa to prevent peripheral conversion (which would cause nausea and reduce CNS delivery).

2.3 Seizures

MECHANISM A seizure is a sudden, abnormal, **excessive** and **synchronous** electrical discharge of cortical neurons. The clinical manifestation depends on which area of cortex is firing.

Type	Features
Generalized tonic-clonic ("grand mal")	Sudden loss of consciousness → tonic (stiffening) → clonic (rhythmic jerking) → postictal confusion. May have tongue biting, incontinence.
Absence ("petit mal")	Brief staring spells (5–10 sec), no postictal phase. Children. EEG: 3-Hz spike-and-wave.
Focal aware ("simple partial")	Localized motor or sensory phenomenon, awareness preserved.
Focal impaired awareness ("complex partial")	Often temporal lobe; automatisms (lip-smacking), staring, postictal confusion.
Status epilepticus	Seizure > 5 min OR repeated without recovery between. Medical emergency.

PATHOLOGY Common seizure triggers: fever (febrile seizures in children), hypoglycemia, hyponatremia, alcohol withdrawal, illicit drugs (cocaine, methamphetamine), sleep deprivation, head injury, stroke, brain tumor, infection, missed antiepileptic doses.

2.4 Increased ICP — How It Injures Neurons

MECHANISM Rising ICP injures neurons by **direct compression** and by **reducing cerebral perfusion pressure (CPP)**:

CPP = MAP – ICP

As ICP climbs toward MAP, CPP falls toward zero — and neurons are starved of blood flow even though large vessels are still patent. Sustained low CPP causes ischemic neuronal death; severe ICP elevation causes herniation and brainstem death.

★ **EXAM TIP** Recall the Cushing triad from Week 4: **HTN + bradycardia + irregular respirations** = late ICP elevation. Restlessness, confusion, behavior change come earlier.

Practice Questions — CNS Disorders

Q5. Select all that apply. Which findings are commonly associated with multiple sclerosis?

- A. Optic neuritis with painful vision loss
- B. Spasticity and hyperreflexia
- C. Sensory disturbances and numbness
- D. Heat intolerance (Uhthoff phenomenon)
- E. Mid-adult-onset chorea
- F. Fatigue and bladder dysfunction

[Show answer](#)

A, B, C, D, F. E (chorea) is Huntington disease. MS produces UMN signs from CNS demyelination, and symptoms classically worsen with heat (Uhthoff).

Q6. A patient with Parkinson's has resting tremor and progressive difficulty initiating movement. Which structural and biochemical change is responsible?

- A. Loss of acetylcholine in the cortex
- B. Loss of dopamine-producing neurons in the substantia nigra
- C. Loss of GABA in the basal ganglia
- D. Loss of serotonin in the brainstem

[Show answer](#)

B. Substantia nigra dopamine loss is the defining lesion of Parkinson's. Levodopa-carbidopa replaces the missing transmitter.

Q7. A patient develops sudden loss of awareness with rhythmic jerking of all four extremities, then becomes confused for 20 minutes. Which type of seizure?

- A. Absence seizure
- B. Focal aware seizure
- C. Generalized tonic-clonic seizure
- D. Myoclonic seizure

[Show answer](#)

C. Loss of consciousness + rhythmic jerking + postictal confusion = tonic-clonic. Absence has no postictal phase. Focal aware preserves awareness.

Q8. Which mechanism most directly causes neuron injury when intracranial pressure rises?

- A. Increased oxygen delivery to neurons
- B. Reduced cerebral perfusion pressure → ischemia

- C. Excess CSF nutrition of neurons
- D. Increased lymphatic drainage

Show answer

B. CPP = MAP – ICP. As ICP climbs, perfusion falls. Neurons that don't receive blood die. Direct compression also contributes, especially with herniation.

Unit 3 • Spinal Cord & Peripheral Nerve Disorders LO1

3.1 Spinal Cord Injury — Phases & Sequelae

Acute trauma → primary mechanical damage → secondary injury (edema, hemorrhage, ischemia, glutamate excitotoxicity, free radicals) → spinal shock (areflexia) → reflex return → spasticity (chronic UMN-type)

MECHANISM SCI involves a primary injury (mechanical) and a more damaging **secondary injury cascade**: vasogenic edema, hemorrhage, ischemia, calcium-mediated cell death, and inflammation. This secondary phase enlarges the lesion over hours to days — which is why minimizing inflammation and maintaining perfusion matters in early management.

WHY Why SCI recovery is slow and incomplete. CNS axons do not regenerate effectively. Adult neurons have limited proliferative capacity, scar tissue (gliosis) blocks regrowth, and inhibitory molecules in the CNS environment actively prevent axon extension. PNS injuries can recover; CNS ones largely cannot.

3.2 Spinal Shock vs Neurogenic Shock vs Autonomic Dysreflexia

	Spinal shock	Neurogenic shock	Autonomic dysreflexia
Timing	Immediately after SCI; lasts days to weeks	Immediately after high SCI (T6 or above)	Weeks to years after SCI at T6 or above
Mechanism	Loss of all reflex activity below the lesion (transient)	Loss of sympathetic tone below lesion → unopposed parasympathetic	Painful or noxious stimulus below the lesion (full bladder, fecal impaction) triggers massive sympathetic surge below; reflex parasympathetic response above
Findings	Flaccid paralysis, areflexia, no autonomic compromise required	Hypotension, bradycardia, warm/dry skin below lesion	Severe HTN, pounding headache, bradycardia, flushing/sweating above lesion, pale/cool below
Treatment	Supportive; reflexes return on their own	Fluids + vasopressors; atropine for bradycardia	Find and remove the stimulus immediately (catheterize, disimpact); raise head; antihypertensive

PATHOLOGY Neurogenic shock is the only "warm shock" of the bunch. Hypovolemic, cardiogenic, and obstructive shock all show *cold* skin from compensatory vasoconstriction. Neurogenic shock has lost sympathetic vasoconstriction — vessels dilate, skin is warm and dry, BP and HR are both LOW (vs other shocks where HR is high).

⚠ **COMMON MISSTEP** "All shock has tachycardia." No — neurogenic shock has bradycardia (loss of sympathetic input to the heart). This is a frequent test trap.

3.3 Disk Herniation & Radiculopathy

MECHANISM The intervertebral disk has a tough outer ring (annulus fibrosus) and a gel-like center (nucleus pulposus). Lifting injuries or aging can cause the nucleus to herniate posteriorly through the annulus, compressing a spinal nerve root → **radiculopathy** — pain, numbness, weakness in that nerve's distribution.

The classic lumbar disk herniation: L4-L5 or L5-S1 → nerve root compression → "sciatica" — back pain radiating down the back of the leg, often into the foot, with dermatomal sensory loss and possibly weakness.

★ **EXAM TIP** **Cauda equina syndrome** is a surgical emergency: large central lumbar herniation compresses the cauda equina. Look for: saddle anesthesia, urinary retention/incontinence, bilateral leg weakness, loss of anal sphincter tone. Decompress within 24–48 h.

Practice Questions — Spinal Cord & PNS

Q9. Order. Place the events of secondary spinal cord injury in correct sequence:

- Free radical generation and inflammation worsen tissue damage
- Mechanical trauma to the spinal cord
- Calcium influx activates destructive enzymes
- Local ischemia and hemorrhage develop
- Cells fail to maintain ion gradients and depolarize

[Show answer](#)

Correct order: Mechanical trauma → ischemia and hemorrhage → cells lose ion gradients and depolarize → Ca^{2+} influx activates enzymes → free radical generation and inflammation. Same energy-failure cascade as stroke, played out in cord tissue.

Q10. A patient with a C5 spinal cord injury has BP 78/45, HR 52, warm dry skin, and paralysis below the lesion. Which best explains this?

- A. Hypovolemic shock from blood loss
- B. Cardiogenic shock
- C. Neurogenic shock from disrupted sympathetic outflow
- D. Septic shock

[Show answer](#)

C. High SCI knocks out sympathetic tone → vasodilation → hypotension, warm skin, and bradycardia (no sympathetic input to the heart). Treat with fluids and vasopressors; atropine for bradycardia.

Q11. A patient asks why recovery from spinal cord injury is often slow and incomplete. Best response:

- A. "Spinal nerves regenerate quickly within days."
- B. "Adult CNS neurons have limited regenerative capacity, and scar tissue plus inhibitory factors prevent axon regrowth."
- C. "Recovery depends on diet alone."
- D. "Recovery is complete within 2 weeks for most patients."

[Show answer](#)

B. CNS axons cannot regrow effectively because of intrinsic neuron limitations, glial scar formation, and inhibitory molecules in the CNS environment. PNS nerves can regrow; CNS largely cannot.

Q12. A patient lifts a heavy box and develops back pain that radiates down the back of one leg with numbness in a dermatomal pattern. Which mechanism best explains the leg symptoms?

- A. Stroke in the spinal cord
- B. Compression of a lumbar nerve root by herniated disk material
- C. Cerebellar lesion
- D. Peripheral neuropathy from diabetes

[Show answer](#)

B. Lumbar disk herniation → nerve root compression → radiculopathy in that root's distribution. This is sciatica — a textbook presentation.

Unit 4 • Meningitis LO1

MECHANISM Meningitis = inflammation of the meninges (especially arachnoid and pia) and the CSF in the subarachnoid space. Most often infectious (bacterial, viral, fungal). Inflammation drives the classic clinical syndrome: **fever, headache, neck stiffness** (nuchal rigidity).

Classic findings

- **Fever** — almost universal
- **Severe headache** — diffuse, worsening
- **Nuchal rigidity** — patient cannot touch chin to chest
- **Photophobia**
- **Altered mental status / decreased alertness**
- **Kernig sign** — pain with knee extension when hip is flexed
- **Brudzinski sign** — hip/knee flex involuntarily when neck is flexed
- **Petechial / purpuric rash** in meningococcal disease

CSF analysis — bacterial vs viral vs fungal

	Bacterial	Viral	Fungal/TB
Appearance	Cloudy/turbid	Clear	Clear or slightly cloudy
WBC predominance	Neutrophils ↑↑	Lymphocytes ↑	Lymphocytes ↑
Glucose	Low (bacteria consume it)	Normal	Low

Protein	↑↑	Normal or mildly ↑	↑
Opening pressure	↑	Normal	↑

★ **EXAM TIP** Bacterial meningitis is a medical emergency. Antibiotics within 1 hour. Don't wait for LP results to start treatment if clinical suspicion is high — give empirical IV antibiotics + dexamethasone, then image and LP.

PATHOLOGY Common pathogens by age:

- **Newborn (< 1 mo):** Group B Strep, E. coli, Listeria
- **Infants/children:** Streptococcus pneumoniae, Neisseria meningitidis, Haemophilus influenzae type b (Hib — much less common now thanks to vaccine)
- **Young adults / college:** N. meningitidis (dorm outbreaks)
- **Older adults:** S. pneumoniae, Listeria

Practice Questions — Meningitis

Q13. A patient develops fever, severe headache, neck stiffness, and decreased alertness. CSF shows ↑ WBC with neutrophil predominance, ↓ glucose, and ↑ protein. Which diagnosis?

- A. Viral meningitis
- B. Fungal meningitis
- C. Bacterial meningitis
- D. Migraine

[Show answer](#)

C. Neutrophilic CSF + low glucose + high protein = bacterial meningitis. Start empirical antibiotics immediately.

Q14. Select all that apply. Which findings are consistent with meningitis in a child?

- A. Fever
- B. Severe headache
- C. Nuchal rigidity
- D. Photophobia
- E. Increased deep tendon reflexes only on one side
- F. Petechial rash (meningococcal)

[Show answer](#)

A, B, C, D, F. E (unilateral hyperreflexia) is more typical of focal CNS lesions like stroke. The other five are classic meningitis findings.

Unit 5 • Neuromuscular Junction (NMJ) LO2

5.1 NMJ Anatomy & Function

Action potential reaches motor neuron terminal → voltage-gated Ca^{2+} channels open → Ca^{2+} entry triggers vesicle fusion → **acetylcholine (ACh)** released into synaptic cleft → ACh binds **nicotinic ACh**

receptors on muscle → Na⁺ influx → muscle depolarization → contraction

MECHANISM Acetylcholinesterase in the synaptic cleft rapidly degrades ACh, ending the signal. This allows precise control of muscle contraction. Drugs that *block* acetylcholinesterase (like pyridostigmine) leave more ACh in the cleft and improve transmission — useful in myasthenia gravis.

Step at NMJ	What goes wrong → disease
Ca ²⁺ entry into nerve terminal	Lambert-Eaton syndrome (autoimmune; antibodies vs voltage-gated Ca ²⁺ channel)
ACh release	Botulism (botulinum toxin blocks ACh release)
Postsynaptic ACh receptor	Myasthenia gravis (autoantibodies block AChR)
Acetylcholinesterase	Organophosphate poisoning (inhibits enzyme → ACh excess → cholinergic crisis)

5.2 Myasthenia Gravis (MG)

MECHANISM Autoimmune disease — IgG antibodies bind and block **acetylcholine receptors** on the postsynaptic muscle membrane. The released ACh has nowhere to bind (or fewer functional receptors), so each muscle action potential is weaker. With repeated activity, ACh stores deplete and weakness progresses; rest restores transmission. This is type II hypersensitivity (receptor blockade).

Hallmark presentation:

- **Fatigable muscle weakness** — worsens with use, improves with rest
- **Ptos** (drooping eyelids) and **diplopia** (double vision) — often the first symptoms
- Weak chewing, swallowing, speech (bulbar muscles)
- Generalized weakness — proximal > distal; respiratory muscles in severe disease (myasthenic crisis)

★ **EXAM TIP "Patient brushes hair or holds objects overhead and gets weaker; rest helps"** — that exam vignette is myasthenia gravis. Confirm with anti-AChR antibodies, edrophonium (Tensilon) test, EMG (decremental response).

PATHOLOGY MG association with the thymus. About 75% of MG patients have thymic abnormalities (hyperplasia or thymoma). Thymectomy can improve symptoms. Treatments: pyridostigmine (acetylcholinesterase inhibitor), corticosteroids, immunosuppressants, IVIG/plasmapheresis for crisis.

WHY Why pyridostigmine helps. By blocking acetylcholinesterase, pyridostigmine slows ACh breakdown — more ACh stays in the cleft, raising the chance of binding the (limited) functional receptors. It treats symptoms; it doesn't fix the autoimmunity.

5.3 Botulism

MECHANISM Botulinum toxin (from *Clostridium botulinum*) is one of the most potent toxins known. It enters motor nerve terminals and cleaves the SNARE proteins required for ACh

vesicle fusion → **blocks ACh release**. No ACh in the cleft → no muscle activation → flaccid paralysis.

Classic presentation:

- Sources: improperly canned/preserved foods (adult botulism), honey in infants < 1 year (infant botulism), wound contamination
- **Descending** symmetric flaccid paralysis (opposite of GBS, which ascends)
- Cranial nerves first: **blurred vision, diplopia, ptosis, dysphagia, dysarthria**
- Progresses to respiratory muscle weakness — **respiratory failure is the killer**
- No sensory loss; mental status preserved

⚠ **COMMON MISSTEP** "Botulism causes spasms." Wrong — botulism causes **flaccid** paralysis (no ACh = no muscle activation). Tetanus, by contrast, causes *spasms* (it blocks inhibitory neurotransmitters in the spinal cord).

ACRONYM / MNEMONIC "4 D's of botulism" (cranial nerves): Diplopia, Dysphagia, Dysarthria, Dysphonia. Plus **D**escending paralysis.

Practice Questions — NMJ

Q15. A patient has fluctuating muscle weakness that worsens with activity and improves with rest. Diagnostic testing reveals autoantibodies. Which mechanism best explains the symptoms?

- A. Antibodies block postsynaptic acetylcholine receptors at the NMJ
- B. Antibodies destroy CNS myelin
- C. Toxin prevents ACh release
- D. Mutation alters dystrophin protein

Show answer

A. Myasthenia gravis. Autoantibodies block AChR → fewer functional receptors → fatigable weakness. Rest replenishes ACh and partial transmission resumes.

Q16. After eating improperly canned food, a patient develops blurred vision, dysphagia, and progressive descending muscle weakness. The toxin prevents the release of acetylcholine. Which diagnosis?

- A. Myasthenia gravis
- B. Botulism
- C. Guillain-Barré syndrome
- D. Multiple sclerosis

Show answer

B. Descending flaccid paralysis + canned food + cranial nerve symptoms first = botulism. Treatment: antitoxin + supportive care including ventilation.

Q17. A patient with myasthenia gravis is prescribed pyridostigmine. The drug improves muscle strength because it:

- A. Increases ACh production in the nerve terminal
- B. Inhibits acetylcholinesterase, leaving more ACh in the synaptic cleft
- C. Blocks ACh receptors directly
- D. Removes anti-AChR antibodies

[Show answer](#)

B. AChE inhibitors prolong ACh activity in the cleft, raising the chance it binds the still-functional receptors. They treat symptoms; immunosuppression treats the underlying autoimmunity.

Q18. Matching. Match each NMJ change to its effect.

Change	Effect
Botulinum toxin	___
Acetylcholinesterase inhibitor	___
Anti-AChR antibody	___
Voltage-gated Ca ²⁺ channel antibody (Lambert-Eaton)	___

Choices: ↓ ACh release at nerve terminal · Blocks postsynaptic receptors · Prevents ACh breakdown · Blocks Ca²⁺ entry → ↓ ACh release

[Show answer](#)

Botulinum → ↓ ACh release · **AChE inhibitor** → prevents ACh breakdown · **Anti-AChR** → blocks receptors · **Lambert-Eaton** → blocks Ca²⁺ entry → ↓ release.

Unit 6 • Pediatric Neurologic Disorders LO3

6.1 Hydrocephalus in Children

MECHANISM Excess CSF accumulation in the ventricles. In infants (skull bones not yet fused), this enlarges the head; in older children with closed sutures, it raises ICP.

Communicating = absorption fails (post-meningitis, post-bleed, congenital).

Non-communicating (obstructive) = flow blocked, often at the cerebral aqueduct (aqueductal stenosis is the most common congenital cause).

Classic findings in infant hydrocephalus:

- **Rapid head growth** — head circumference crosses percentiles
- **Bulging fontanelle** with widened sutures
- **"Sunset eyes"** — downward eye deviation from pressure on the upper midbrain
- Poor feeding, vomiting, irritability or lethargy
- High-pitched cry, scalp vein engorgement

PATHOLOGY Untreated hydrocephalus causes permanent brain injury by compressing brain tissue against the skull and reducing perfusion. Even if CSF pressure is later relieved, the damaged white matter and cortex don't recover. **Early shunt placement** (VP shunt) prevents progression.

★ **EXAM TIP** Late ICP signs in young children — vomiting, irritability, bradycardia, irregular breathing, decreased alertness — should prompt urgent imaging, not just a "wait and see." This is the pediatric Cushing-triad analog.

6.2 Neural Tube Defects

MECHANISM The neural tube closes between weeks **3 and 4** of embryonic development. Failure of closure produces a spectrum of defects depending on level and severity.

Defect	Description	Severity
Spina bifida occulta	Vertebral arch fails to fuse; spinal cord and meninges intact; tuft of hair or dimple over the area	Often asymptomatic
Meningocele	Meninges protrude through the vertebral defect; spinal cord intact	Mild to moderate
Myelomeningocele	Meninges and spinal cord protrude → motor/sensory deficits, bowel/bladder dysfunction, frequently associated with hydrocephalus and Chiari II malformation	Severe
Anencephaly	Failure of cranial neural tube closure; absent brain development	Incompatible with life

WHY Folic acid prevention. Adequate folate before conception and through the first trimester reduces neural tube defects by 50–70%. Recommendation: 400 mcg daily for women of childbearing age; 4 mg daily for women with a prior affected pregnancy. The critical window is the first month — often *before* a woman knows she's pregnant — which is why supplementation is recommended for all women who could become pregnant.

6.3 Cerebral Palsy

MECHANISM Non-progressive motor disorder caused by injury to the developing brain. Causes include prematurity, hypoxic-ischemic injury at birth, intrauterine infection, kernicterus, postnatal stroke or trauma, and congenital brain malformation.

Cardinal features:

- Motor impairment present from infancy/early childhood
- Most common type: **spastic** (UMN signs — increased tone, hyperreflexia, scissoring gait)
- Other types: dyskinetic (chorea, athetosis), ataxic (cerebellar)
- Often associated with intellectual disability, seizures, vision/hearing problems, learning disorders — but cognitive function may also be entirely normal
- **Non-progressive** — the brain injury itself doesn't worsen, though musculoskeletal problems may evolve with growth

⚠ **COMMON MISSTEP "Cerebral palsy is progressive."** No — the brain injury is fixed (occurred prenatally, perinatally, or in early childhood). The clinical picture may evolve as the child grows, but the lesion itself is static. Contrast with neurodegenerative diseases.

6.4 Pediatric Meningitis

PATHOLOGY Children with meningitis present similarly to adults: **fever, headache, vomiting, neck stiffness, photophobia**. CSF analysis is the same (neutrophil-predominant low-glucose CSF in bacterial). Petechial rash strongly suggests

meningococcal disease — call infection control. Vaccines (pneumococcal, Hib, meningococcal) have dramatically reduced incidence.

★ **EXAM TIP** In **infants**, the classic neck stiffness is often absent. Look for: poor feeding, lethargy, irritability, bulging fontanelle, fever or hypothermia, seizures. A high index of suspicion is needed.

Practice Questions — Pediatric Neuro

Q19. An infant has rapid head growth, poor feeding, a bulging fontanelle, and downward eye deviation. Most likely diagnosis?

- A. Cerebral palsy
- B. Hydrocephalus
- C. Spina bifida occulta
- D. Febrile seizure

[Show answer](#)

B. Bulging fontanelle + sunset eyes + rapid head growth = infant hydrocephalus. Imaging confirms; VP shunt is the typical treatment.

Q20. Order. Place the events of obstructive hydrocephalus in correct sequence:

- ICP rises and brain tissue is compressed
- Lateral and third ventricles enlarge
- CSF flow is blocked at the cerebral aqueduct
- CSF cannot reach the fourth ventricle for absorption

[Show answer](#)

Correct order: CSF flow blocked at aqueduct → CSF cannot reach 4th ventricle → lateral and 3rd ventricles enlarge → ICP rises and brain tissue is compressed.

Q21. A pregnant patient asks when neural tube defects occur during development. Best response:

- A. "During the third trimester."
- B. "Before week 3 of pregnancy, neural tube defects do not occur."
- C. "Around weeks 3–4 of embryonic development, when the neural tube normally closes."
- D. "Only after exposure to certain medications late in pregnancy."

[Show answer](#)

C. Neural tube closure happens between weeks 3 and 4. This is why folic acid supplementation is recommended *before* conception — by the time most women know they're pregnant, the critical window has passed.

Q22. Select all that apply. Which findings are consistent with cerebral palsy?

- A. Motor impairment present from early childhood
- B. Spasticity and hyperreflexia in the most common subtype
- C. Progressive worsening of the underlying brain lesion
- D. Often associated with seizures and learning disorders
- E. Caused by intrauterine infection, perinatal hypoxia, or prematurity

[Show answer](#)

A, B, D, E. C is wrong — CP is **non-progressive**. The brain injury is fixed; the clinical presentation may evolve with growth but the lesion does not worsen.

Q23. Matching. Match each pediatric condition to its main mechanism.

Condition	Mechanism
Hydrocephalus	___
Spina bifida	___
Cerebral palsy	___
Bacterial meningitis	___

Choices: Failed neural tube closure · CSF accumulation in ventricles · Bacterial inflammation of meninges · Non-progressive brain injury affecting motor function

[Show answer](#)

Hydrocephalus → CSF accumulation in ventricles · Spina bifida → failed neural tube closure · Cerebral palsy → non-progressive brain injury · Bacterial meningitis → bacterial inflammation of meninges.

Q24. A young child has vomiting, irritability, bradycardia, irregular breathing, and enlarged ventricles on imaging. Which best explains the bradycardia and breathing pattern?

- A. Beginning sepsis
- B. Late ICP elevation (Cushing reflex equivalent) from hydrocephalus
- C. Allergic reaction
- D. Hyperglycemia

[Show answer](#)

B. The triad of bradycardia + irregular respirations + altered mental status, in the setting of enlarged ventricles, indicates dangerous ICP elevation. Urgent decompression is needed.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 5 Missteps

#	The trap	What's actually true
1	"All strokes get tPA."	Only ischemic, within 3–4.5 h of last known well, hemorrhage excluded, no contraindications.
2	"Atrial fibrillation causes hemorrhagic stroke."	AFib → embolic ischemic stroke (clot from atrium).
3	"Neurogenic shock is hot and tachycardic."	It's hot but bradycardic — sympathetic outflow lost.
4	"Spinal shock is permanent."	Temporary phase. Spasticity follows as reflex arcs reorganize.
5		It's PNS — root compression below the cord level.

	"Disk herniation is a brain problem."	
6	"Bacterial meningitis CSF has lymphocytes."	Bacterial = neutrophils . Lymphocytes = viral or fungal/TB.
7	"Myasthenia gravis is muscle weakness from low calcium."	It's autoantibodies blocking AChR at the NMJ. Type II hypersensitivity.
8	"Botulism causes spasms."	Botulism = flaccid , descending paralysis. Tetanus causes spasms.
9	"GBS and botulism are the same."	GBS ascends; botulism descends. GBS is autoimmune demyelination; botulism blocks ACh release at NMJ.
10	"Cerebral palsy is progressive."	Non-progressive brain injury from before/around birth.
11	"Folic acid late in pregnancy prevents neural tube defects."	Closure is at weeks 3–4. Folic acid must be on board before conception.
12	"Hydrocephalus = too much CSF production."	Usually impaired flow or absorption, not overproduction.
13	"Pyridostigmine cures myasthenia."	It treats <i>symptoms</i> by inhibiting AChE. Doesn't fix the autoimmunity.
14	"Sciatica is a CNS disorder."	It's a radiculopathy from PNS root compression.
15	"Infant meningitis presents with neck stiffness."	Classic neck stiffness is often absent in infants. Look for irritability, bulging fontanelle, poor feeding, fever or hypothermia.

Master Mnemonic Sheet

Mnemonic	Meaning
FAST	Stroke recognition: Face, Arm, Speech, Time
"Time is brain"	~2 million neurons die per minute in stroke
TRAP	Parkinson's: Tremor, Rigidity, Akinesia, Postural instability
"Ascending = GBS, Descending = Botulism, Fatigable = MG"	Three flaccid paralyses by direction/pattern
4 D's of botulism	Diplopia, Dysphagia, Dysarthria, Dysphonia + Descending
CPP = MAP – ICP	As ICP rises, perfusion falls → ischemia
"Warm shock = neurogenic"	Loss of sympathetic vasoconstriction; the only "warm" shock
"Bacterial = cloudy + low glucose + neutrophils"	CSF profile of bacterial meningitis
Sunset eyes + bulging fontanelle	Infant hydrocephalus
"Folate before pregnancy"	Neural tube closes at weeks 3–4
"Static, not progressive"	Cerebral palsy

Anti-AChR = MG	Diagnostic antibody for myasthenia gravis
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NMJ Disease Quick Reference

Disease	Defect	Direction of weakness	Hallmark
Myasthenia gravis	Anti-AChR (postsynaptic block)	Fluctuates, fatigable	Ptosis, diplopia, fatigable weakness improved by rest
Botulism	Toxin blocks ACh release	Descending, flaccid	4 D's after canned food / wound / infant honey
Lambert-Eaton	Anti-VGCC (presynaptic)	Improves with use	Often paraneoplastic (small-cell lung cancer)
Organophosphate poisoning	Inhibits AChE → ACh excess	Cholinergic crisis	SLUDGE (Salivation, Lacrimation, Urination, Defecation, GI distress, Emesis)

Week 5 Study Strategy

- **Lock down stroke first.** Ischemic vs hemorrhagic mechanism, FAST, tPA window, AFib → embolism — all guaranteed test points.
- **Memorize the SCI shocks.** Spinal shock (areflexia phase), neurogenic shock (warm + brady), autonomic dysreflexia (massive HTN above lesion). Each has a different timeline and management.
- **Learn the meningitis CSF profile cold.** Bacterial vs viral vs fungal — neutrophils vs lymphocytes, glucose, protein.
- **Drill the NMJ direction-of-weakness pattern.** MG fatigable, botulism descending, GBS ascending. These three trip students up regularly.
- **For pediatric topics, connect timing to mechanism.** Neural tube defects = weeks 3–4. CP = peri/postnatal injury. Hydrocephalus = any age, depending on cause.
- **"Static vs progressive"** — cerebral palsy is the textbook static lesion. Don't miss it.
- **Recognize folic-acid-before-conception teaching.** Frequent SATA distractor.

SCI225 Week 5 Study Guide — built from the Week 5 learning objectives covering CNS and PNS disorders, the neuromuscular junction, and pediatric neurologic alterations. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Same color-coded format as previous weekly guides. Print for paper, or use in-browser with collapsible practice answers.